

Predicting Quebec Winter Electricity Demand

Abstract

Our project aims to examine whether a machine learning model can accurately predict hourly electricity consumption in Quebec during winter. Our research will use the publicly available Hydro-Quebec database on electricity consumption by customers participating in the Hilo demand-response program. This database includes more than 64,000 hourly entries between January 2022 and June 2024 from three anonymous substations in Montreal, with information on electricity consumption, weather parameters such as wind, humidity, and snow, and the number of customers. In addition, we plan to use Statistics Canada's electricity production and availability data for Quebec from January 2008 to July 2026.

Our Goal is to predict hourly electricity consumption for each substation. We will first clean the data, then split it chronologically into training, validation, and test sets to ensure information from the future isn't leaked into the training data. We plan on training a linear regression model as well as a decision tree model. Both of these models will be compared against a simple baseline using mean absolute error, root mean squared error, and mean absolute percentage error. In the end, we will report the results overall and separately for normal hours and Hilo demand-response hours.

Research Background

Most homes in Quebec heat with electricity, and as a result, demand rises especially during the cold winters. Those peaks are the most expensive times for Hydro-Québec to supply. To combat this, Hydro-Québec runs demand-response programs such as Hilo, which pre-heat homes before a peak event and then lower their thermostat settings during the event.

Accurate forecasting of electricity use in advance could help Hydro-Québec decide when to run these events and estimate how much electricity demand they could reduce. However, during the winter, this becomes especially challenging because usage is affected by several factors, including weather, time of day, and customer behaviour.

Our goal is to determine whether a machine learning model can predict these patterns and identify the factors most useful for predicting electricity consumption.

Methodology

Dataset

Our main dataset comes from Hydro-Québec's open dataset, Electricity consumption of customers taking part in a local demand response program. It contains over 64,000 hourly records and 30 columns from three anonymized substations in Montreal from January 2022 to June 2024. The output variable is the energy consumed each hour, and the input features include weather, the number of customers, time of day, data, smart thermostats and whether a Hilo peak-reduction event was taking place.

Our second dataset comes from Statistics Canada's Table 25-10-0016-01 Electric power generation, monthly receipts, deliveries and availability. This dataset contains monthly electricity data for each

province from January 2008 to July 2026. This will help us get a broader view of Quebec's seasonal electricity demand and check whether the patterns in the substation data make sense.

Data Processing

We will clean the data by filling in short gaps in missing hours and removing some extreme readings that appear to be meter errors. We will also add useful features, such as consumption for the previous hour and day. Instead of splitting the data randomly, we will split it by time so the models are trained on earlier data and tested on later data.

Models and Methods

We plan to use two supervised learning methods, linear regression and a decision tree. Linear regression will help give us an easy-to-explain model of how consumption changes with temperature, time of day, and peak event. A decision tree can be used to capture more complex-looking patterns, such as consumption rising faster on very cold days. Both of these will be compared against a baseline that predicts each hour's consumption using the same hour from the previous day.

Evaluation

We will compare the models using mean absolute error, root mean square error, and mean absolute percentage error. We will also check how the models perform during peak hours and Hilo Events.

Significance

Winters can cause Hydro-Québec to incur higher costs, increase imports, and face reliability risks. Our project will examine how well standard machine learning models perform in forecasting hourly consumption during these high-demand periods. We will also assess whether the model can maintain accuracy during a demand-response event. The feature analysis will also show which factors, such as temperature or time of day, are most useful for planning a peak-reduction program.

SMART Timeline

Milestone	Activities	Deliverable
M1 - Fri Sep 18 (revised)	Finalize proposal and team contract; create the shared repository; all members review the dataset	Revised proposal PDF + contract + repository link
M2 - Fri Oct 2	Select a dataset, run EDA, define the target, splits, metrics, and risks	Data card + EDA notebook + defined target
M3 - Mon Oct 19	Build the data pipeline, implement the baselines, and make results reproducible	Working pipeline + linear regression baseline with a results table
M4 - Fri Oct 30	Review methods and results,	Decision tree added +

	identify blockers, update timeline	comparison table + progress slides presentation
M5 - Fri Nov 20	Train and compare both models, analyze errors, and draft conclusions	Compare linear regression and decision tree model + draft results and conclusions
M6 - Fri Dec 4	Finalize results, document contributions	Finalize results + report + repository + presentation demo
M7 - Mon Dec 7	Each member reflects on their own contribution	Individual reflection PDF

References

Electricity consumption of customers taking part in a local demand response program. (2024, November 27).

<https://donnees.hydroquebec.com/explore/dataset/consommation-clients-evenements-pointe/information/>

Government of Canada, Statistics Canada. (2026, September 29). Electric power generation, monthly receipts, deliveries and availability. <https://www150.statcan.gc.ca/t1/tb11/en/tv.action?pid=2510001601>

Peak demand events – Winter. (2026, October 2).

<https://donnees.hydroquebec.com/explore/dataset/evenements-pointe/table/?disjunctive.offre&disjunctive.plagehoraire&disjunctive.secteurclient&sort=datedebut>